

Joud Mawad

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judemawad.com

Education

Technische Universität Braunschweig 2023–Present

B.Sc. Computer Science

- Relevant coursework: Software Development Lab, Software Engineering, Programming I & II, Algorithms and Data Structures I & II, Databases, Computer Networks, and Computer Engineering.

Projects

Ignite: Fitness & Nutrition 2023–2024

- Built and published a native iOS app for nutrition, training, hydration, and progress tracking using Swift, SwiftUI, Core Data, and HealthKit.
- Integrated Open Food Facts and REST APIs for food lookup and barcode scanning, with local-first personal data storage and Supabase-backed user submissions.

Voice-Controlled LED Cube (In Progress) 2026–Present

- Built the core system around a Raspberry Pi 5, 64×64 HUB75 LED matrix, ReSpeaker microphone array, speaker output, and Linux services for audio and startup control.
- Implemented a local AI assistant with tool-based actions, wake-word detection, local Whisper speech recognition, text-to-speech, and custom LED visualizations.
- Integrated the software with the physical device and prototyped the enclosure, mounts, cooling, and internal layout for 3D printing.

Home Lab & Self-Hosted Infrastructure 2025–Present

- Set up and maintain an Ubuntu server with Docker Compose, remote access, and containerized services including Nextcloud, Jellyfin, and Navidrome.
- Troubleshoot networking, storage, containers, updates, and service availability while documenting configurations and maintenance steps.

ClockWise: Web Time Tracking 2025

- Developed and tested front-end components with HTML, CSS, and JavaScript in a TU Braunschweig software-development team.
- Collaborated through Git-based workflows, structured task coordination, and technical documentation.

Portfolio & Project Details: judemawad.com

Skills

Programming: Swift, Python, HTML, CSS, JavaScript, SQL

Mobile & Data: SwiftUI, Core Data, HealthKit, REST APIs, Supabase

Systems & Infrastructure: Linux, Ubuntu Server, Docker, Docker Compose, SSH, Tailscale/WireGuard, networking fundamentals

Tools & Workflow: Git, GitHub, Xcode, VS Code, Jira, Linear

Prototyping: Fusion 360, Blender, 3D printing

Languages: Arabic (native), English (C1), German (C1)